

Universal Indicator Lab — Code & Logic Audit

MarketPil / NSA Intraday Assistant · Audit date 5 Oct 2026 · Branch `merge/market-analysis-testlab` (commit `ca8fea2` + uncommitted direction-conflict patch) · Audit only — no code was changed

Verdict. Every indicator on the screen is backed by a real backend calculation — none is UI-only — and the formulas are standard and causal (no future data). The main problems are that the screen mixes **three separate BUY/SELL engines**, and that several panels and controls do not reflect what the backend actually does.

1. Scope and method

- **Screen:** Universal Indicator (Test) Lab — Test Configuration, Chart & Signals, Indicator Status strip, Current Candle, Final Signal, Market Signal (15M→5M), Indicator Ablation, Live signal cards.
- **Backend:** `backend/testlab/*` (engine, components, `htfEngine`, `decision`, `context15`, `strikeGamma`, `oiValidation`, `optionsData`, `live`, `runner`), `backend/indicators/index.ts`, `backend/liquidity/orderBlock.ts`, `backend/routes/api.ts` (test-lab routes).
- **Method:** read the code path from each screen element to its backend source; checked formulas; checked every calculation for future-candle use; verified data behaviour on stored runs (1 Oct 2026 futures run and the 30-session spot run, 20 Aug–1 Oct 2026); automated test suite (587 tests passing).
- Anything that could not be confirmed is marked **NOT VERIFIED**.

2. What is correct

Area	Finding (verified in code)
EMA 9 / 21	Standard EMA seeded with a simple average — <code>indicators.ema</code> .
ATR 14	True range with Wilder smoothing — <code>indicators.atr</code> .
VWAP	Typical price × volume, resets each session — <code>indicators.vwap</code> . Futures VWAP uses the same futures contract as the signal series.
UT Bot	ATR trailing stop with crossover state — <code>components.utBot</code> .
Linear regression, Volume state	Regression over a trailing window; volume compared with the median of earlier candles, excluding the current one.
Structure / BOS, S/R	3-candle pivots, break confirmed on a close, events sorted by time (<code>orderBlock.detectBOS</code>); support / resistance from swing pivots.
No look-ahead	Every per-candle calculation uses candles up to and including that candle only. 15M context uses only the last <i>closed</i> 15M candle. Replay keeps only candles closed by the replay time (5M and 15M). The forming candle's option bar is hidden. Decision-layer entry = candle close. "Genuine / missed / timing" labels are computed afterwards and never fed back.
15M + 5M integration	The decision layer reuses the existing 15M engine (<code>direction15Series</code>) through <code>context15ForSeries</code> , and the 5M engine's own per-candle values.
R:R	No R:R gate remains in any of the three engines (checked in code and by tests). R:R is information only.
OI / Gamma / Options	OI never blocks the decision layer; OI age shown as LIVE / FRESH / AGING / DELAYED / STALE. Gamma is model-derived (Black-Scholes on the feed IV) and labelled so. Missing option data shows DATA UNAVAILABLE; $IV \leq 0$ is treated as missing.
Ablation	Switch names on screen match the names the engines read.

3. Issues found

#	Severity	File → function	Problem	Impact
1	High	engine.ts → runEngine , htfEngine.ts → runHtfEngine , decision.ts → runDecisionLayer	Three engines produce BUY/SELL for the same candle with different rules. Run Indicator Test, Final Signal and chart markers use runEngine ; the Market Signal tab uses runHtfEngine ; the live cards use runDecisionLayer . On 1 Oct: 1 BUY + 6 SELL vs 4 SELL vs 3 trades.	Conflicting recommendations on one screen.
2	High	app.js → uil2RenderFinalSignal , fed by api.ts → tlShape (lastRow)	"Final Signal" shows only the old engine's last candle. Entry / SL / T1 / T2 appear only if that exact candle is a signal, so the panel usually shows "—". It is not updated in Live mode.	Explains "BUY/SELL not showing Entry / SL / T1 / T2".
3	High	api.ts → tlDecisionShape (no front-end consumer)	The decision layer's test-run results (signals, plans, movement events) are sent to the browser but never displayed.	Test runs show only the old engine.
4	High	engine.ts → runEngine (data-quality check)	With futures data, missing OI marks every candle BLOCKED ("HISTORICAL OI UNAVAILABLE").	Old engine can go all-WAIT because of OI — against the "OI must not block" rule.
5	Medium	index.html toggles; app.js → uil2Config	"Include Open Interest", "Include Expiry Days", "Use Normalized ATR" and the Instrument Type dropdown are never read. Instrument Type says "Index (Signal)" while the signal series is futures when "Use Futures VWAP" is on.	Controls that do nothing; wrong label.
6	Medium	app.js → uil2IndList	Indicator tile is still named " R:R Gate " with pass/fail against 2.0; shows "—" on WAIT candles.	Misleading — R:R no longer gates.
7	Medium	engine.ts / htfEngine.ts (SL / target)	SL = old swing high + 0.5 ATR; T1 = nearest swing low. Market Signal tab shows R:R 0.02–0.17 on 1 Oct. The decision layer uses a different SL / target formula.	Two different SL / targets for the same candle; old-engine levels are not practical.
8	Medium	engine.ts / htfEngine.ts	Uses the next candle's open to choose T1 and compute risk at decision time.	Small look-ahead in level choice (not in direction).
9	Medium	live.ts → liveDecision / fillStrike	Gamma on the FORMING card is always UNAVAILABLE: the forming option bar is hidden on purpose and the live chain gives only total-day volume.	Gamma missing on the early read.
10	Medium	components.ts → fakeMoveAt , copy in htfEngine.ts , trap check in decision.ts	Fake move is calculated three different ways; the old engine blocks both sides on any trap.	Same candle is "fake" in one engine and not in another.
11	Low	optionsData.ts → deriveExpiry	Option expiry from a weekday rule; holidays handled only for dates already fetched.	Wrong time-to-expiry on a holiday week → model delta / gamma and option SL / target off.
12	Low	htfAudit.ts → perCandleAudit + package README	Hard-coded 14:30 cutoff and "R:R ≥ 2" text in export files (not on screen).	Out-of-date audit files.
13	Low	Ablation (uil2Config , /testlab/htf-run)	Ablation reaches only the old engines; the Market Signal run ignores it.	Ablation results incomplete.

14	Low	context15.ts	"15M context" is computed on 15m / 25m / 60m runs, where it is meaningless.	Misleading on higher timeframes.
15	Low	optionsData.ts (Dhan rolling options)	Rolling closes differ from the listed contract by about ₹0.8 (1 Oct, 41 candles checked).	Small option-price error.

4. Specific checks requested

Check	Result	Detail
Every indicator has real backend logic	Yes	All tiles and panels map to backend values; 3 toggles + Instrument Type are UI-only controls (issue 5).
Formulas correct	Yes	Standard EMA, Wilder ATR, session VWAP, UT Bot, regression, pivots / BOS.
Data source / timeframe	Mostly	5M signal series = current-month futures (or index spot); 15M = same instrument; options = Dhan rolling options ATM $\pm$ 3. Label mismatch (issue 5); 15M context on higher TFs (issue 14).
No look-ahead	Mostly	Decision layer and indicators are causal; old engines use next-candle open for level choice (issue 8).
UI matches backend	Partly	Values shown match their source, but Final Signal / tables come from a different engine than the live cards (issues 1–3).
OI / Gamma / Volume / EMA / VWAP / ATR / Structure implemented	Yes	All implemented in backend.
RSI / MACD implemented	No	Not in the Lab at all (backend or screen). They exist elsewhere in the app only.
Duplicate / conflicting calculations	Yes	Three signal engines (1), two SL / target methods (7), three fake-move definitions (10).
Indicator incorrectly blocks BUY CE / BUY PE	Yes	Missing OI blocks the old engine (4); old fake-move blocks both sides (10). No R:R block anywhere.
Null / stale / missing data	Good	DATA UNAVAILABLE, $IV \leq 0 \rightarrow$ missing, ATR fallback, OI age states, live data age (STALE after 150 s).
Integration with 15M + 5M engine	Yes	Decision layer reuses the existing 15M engine and 5M values; the old engine has no 15M context.

5. Missing

- **RSI and MACD** — not implemented in the Lab.
- **Bid / ask spread** — Dhan does not provide it (historical or live chain).
- **Option Target 2** — not calculated by the existing engine.
- **OI walls** — taken only from ATM $\pm$ 3 strikes, not the full option chain.

6. NOT VERIFIED

- What Dhan's index "volume" actually measures (it is non-zero on index spot: 367 EXPANSION / 1,659 NORMAL / 224 WEAK of 2,250 candles in the 30-session run).
- The "provisional" structure label when no VWAP is passed into `detectMarketStructure`.
- Live behaviour during market hours (only replays were tested).
- Rolling options for monthly-expiry indices (BANKNIFTY, FINNIFTY) and SENSEX.
- A field-by-field match between on-screen tables and export files.

7. Top 10 issues

Issues 1 – 10 in section 3, in that order. Issues 1 – 4 are High severity.

8. Recommended fixes (not implemented)

#	Fix	Addresses
1	Use one engine (the decision layer) for Final Signal, chart markers and tables — or label each panel with the engine behind it.	Issue 1
2	Feed Final Signal and the indicator tiles from the decision layer's latest plan, and update them in Live mode.	Issue 2
3	Add a Lab view for the decision layer's test-run results.	Issue 3
4	Make missing OI a warning in the old engine instead of a block.	Issue 4
5	Connect or remove the three dead toggles and Instrument Type.	Issue 5
6	Rename the tile to "R:R (info)".	Issue 6
7	Use the decision layer's SL / Target in the Market Signal tab and Run Indicator Test (pending owner decision).	Issue 7
8	Use the candle close as the decision entry in the old engines.	Issue 8
9	Show "live gamma: not calculated" explicitly, or build it from successive option-chain snapshots.	Issue 9
10	Use one fake-move definition everywhere.	Issue 10
11	Use the exchange expiry list for holiday weeks; update the stale 14:30 / R:R text in export files.	Issues 11, 12

9. Notes

- Correction to an earlier statement: index spot data is *not* empty of volume / VWAP — Dhan returns index volume and VWAP behaves normally in spot mode.
- The 5M direction-conflict / support-resistance rejection patch is implemented and tested (587 tests passing) but not yet committed at the time of this audit.
- All signals in the Lab are advisory only; no orders are placed (LIVE_ORDER_EXECUTION = false).